

Hematologic Health and Hemostasis

Blood-cell production, oxygen carriage, coagulation, marrow, clonal disease, transfusion, and measurement engineering

Volume 03G of the Health Atlas Research Suite

Purpose

Integrate hematopoiesis, blood-cell function, hemostasis, rheology, clonal evolution, transfusion medicine, and device/assay engineering into a longitudinal whole-person profile.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

This volume is an evidence and engineering synthesis, not a transfusion threshold, anticoagulation protocol, chemotherapy plan or emergency bleeding pathway. Local laboratory methods and clinical context govern interpretation.

How to use this volume

Hematologic health is the capacity to produce and maintain functional blood cells, carry oxygen, defend against infection, stop bleeding without pathologic thrombosis, preserve microvascular flow, and recover from stress or injury. Counts and clotting times are proxies within dynamic systems; interpretation requires preanalytics, morphology, mechanism, organ context, medications, symptoms, function, and trajectory.

Questions this volume is designed to answer

1. How do marrow production, blood-cell function, hemostasis and flow interact?
2. What can counts and coagulation assays establish, and what remains latent?
3. How are bleeding and thrombosis balanced across disease, pregnancy, surgery and drugs?
4. How should point-of-care and continuous blood measurements be validated?
5. Where do molecular classifiers change care, and where do they only add labels?

Scope

- Normal hematopoiesis, blood-cell biology, oxygen carriage, hemostasis and fibrinolysis.
- Anemia, hemoglobinopathy, cytopenia, thrombosis, bleeding, marrow failure and hematologic malignancy.
- Laboratory, imaging, cellular, molecular, bedside and emerging continuous measurement.
- Transfusion, anticoagulation, factor replacement, disease-modifying, cellular and gene therapies.
- Engineering and validation of hematology and coagulation devices.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

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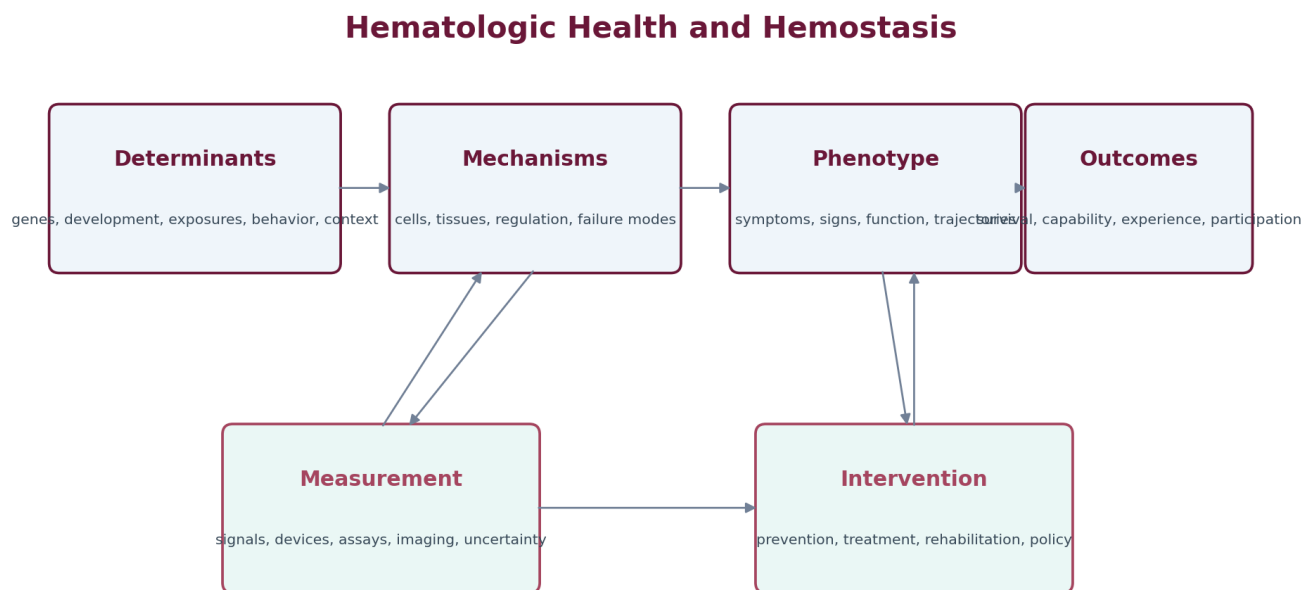
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Executive synthesis

Hematologic health is the capacity to produce and maintain functional blood cells, carry oxygen, defend against infection, stop bleeding without pathologic thrombosis, preserve microvascular flow, and recover from stress or injury. Counts and clotting times are proxies within dynamic systems; interpretation requires preanalytics, morphology, mechanism, organ context, medications, symptoms, function, and trajectory.

Central organizing principle

Integrate hematopoiesis, blood-cell function, hemostasis, rheology, clonal evolution, transfusion medicine, and device/assay engineering into a longitudinal whole-person profile.



A health profile is a causal, temporal, functional, and measurement model - not a single score.

Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.
- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.
- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.

- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

Major bleeding, suspected pulmonary embolism, stroke symptoms, severe anemia with instability, febrile neutropenia, transfusion reaction, thrombotic microangiopathy, or rapidly evolving cytopenia can be emergencies. Experimental blood-contacting devices create infection, thrombosis, hemolysis and electrical hazards and require formal oversight.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Ancient to early modern	Descriptions of pallor, bleeding, fever and blood loss preceded cellular understanding.	Established an empirical vocabulary for hematology while care remained descriptive.	Mechanisms were mixed with humoral, spiritual, and observational explanations.
Seventeenth to nineteenth centuries	Microscopy, circulation, marrow pathology and staining localized cells and diseases.	Connected anatomy and tissue structure to function and disease.	Static specimens could not explain regulation, kinetics, or lived function.
Late nineteenth to mid-twentieth century	Cell counters, blood typing, coagulation assays, component therapy and antibiotics transformed diagnosis and survival.	Made quantities, cell populations, and treatment response measurable.	Reference ranges and categories often hid heterogeneity and preanalytic error.
Late twentieth century	Molecular hematology, cytogenetics, monoclonal antibodies and transplantation linked lesions to treatment.	Linked genes, receptors, signaling pathways, and targeted therapy to phenotype.	Single-pathway explanations frequently underrepresented networks and context.
Twenty-first century	Flow cytometry, deep sequencing, residual-disease assays, engineered cells and gene editing enable high-resolution profiles.	Enabled longitudinal, multimodal, high-dimensional, and precision approaches.	Clinical utility, transportability, cost, equity, and data governance remain limiting.
Current systems model	Hematology is now modeled as a coupled, adaptive system across molecules, cells, organs, behavior, environment, and care.	Supports cross-volume reasoning and explicit measurement models.	Education, reimbursement, datasets, and specialties still reinforce silos.

2.0.1 Descriptions of pallor, bleeding, fever and blood loss preceded cellular understanding.

Scientific consequence. Established an empirical vocabulary for hematology while care remained descriptive.

What remained unresolved. Mechanisms were mixed with humoral, spiritual, and observational explanations.

Evidence anchors: ^[3,6]

2.0.2 Microscopy, circulation, marrow pathology and staining localized cells and diseases.

Scientific consequence. Connected anatomy and tissue structure to function and disease.

What remained unresolved. Static specimens could not explain regulation, kinetics, or lived function.

Evidence anchors: [3,6]

2.0.3 Cell counters, blood typing, coagulation assays, component therapy and antibiotics transformed diagnosis and survival.

Scientific consequence. Made quantities, cell populations, and treatment response measurable.

What remained unresolved. Reference ranges and categories often hid heterogeneity and preanalytic error.

Evidence anchors: [3,6]

2.0.4 Molecular hematology, cytogenetics, monoclonal antibodies and transplantation linked lesions to treatment.

Scientific consequence. Linked genes, receptors, signaling pathways, and targeted therapy to phenotype.

What remained unresolved. Single-pathway explanations frequently underrepresented networks and context.

Evidence anchors: [3,6]

2.0.5 Flow cytometry, deep sequencing, residual-disease assays, engineered cells and gene editing enable high-resolution profiles.

Scientific consequence. Enabled longitudinal, multimodal, high-dimensional, and precision approaches.

What remained unresolved. Clinical utility, transportability, cost, equity, and data governance remain limiting.

Evidence anchors: [3,6]

2.0.6 Hematology is now modeled as a coupled, adaptive system across molecules, cells, organs, behavior, environment, and care.

Scientific consequence. Supports cross-volume reasoning and explicit measurement models.

What remained unresolved. Education, reimbursement, datasets, and specialties still reinforce silos.

Evidence anchors: [1,15]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Hematopoietic stem and progenitor hierarchy

Core principle. Self-renewing stem cells generate erythroid, myeloid, megakaryocytic, lymphoid, and dendritic lineages through regulated differentiation.

Control architecture. Niche signals, transcriptional programs, cytokines, nutrients, oxygen sensing, and feedback from peripheral demand.

Failure modes. Marrow failure, clonal expansion, ineffective maturation, lineage skew, or infiltration.

Observable consequences. Blood counts, reticulocytes, morphology, marrow cellularity, clonality, and lineage-specific function.

Cross-system coupling. Immune defense, oxygen transport, renal erythropoietin, liver thrombopoietin, bone niche, and malignancy.

Evidence anchors: ^[3,5,6]

3.0.2 Erythrocyte production and oxygen carriage

Core principle. Red cells package hemoglobin, deform through microcirculation, and transport oxygen and carbon dioxide over a finite lifespan.

Control architecture. Erythropoietin, iron availability, folate/B12, marrow integrity, hemoglobin synthesis, splenic clearance.

Failure modes. Anemia, erythrocytosis, hemoglobinopathy, membrane/enzyme failure, hemolysis, or impaired deformability.

Observable consequences. Hemoglobin, hematocrit, indices, reticulocytes, smear, hemolysis markers, oxygen content and symptoms under demand.

Cross-system coupling. Respiratory gas exchange, cardiac output, renal sensing, metabolism, exercise, and cerebral perfusion.

Evidence anchors: ^[3,4,7]

3.0.3 Leukocyte production and host defense

Core principle. Granulocytes, monocytes, lymphocytes, and specialized subsets traffic between marrow, blood, tissues, and lymphoid organs.

Control architecture. Growth factors, antigenic signals, circadian release, adhesion, chemokines, and apoptosis.

Failure modes. Neutropenia, leukocytosis, dysfunctional cells, leukemia, immune deficiency, or harmful inflammation.

Observable consequences. Differential count, morphology, flow cytometry, marrow studies, microbial outcomes and inflammatory markers.

Cross-system coupling. Innate/adaptive immunity, infection, autoimmunity, cancer and treatment toxicity.

Evidence anchors: [3,6,9]

3.0.4 Primary hemostasis

Core principle. Platelet adhesion, activation, secretion, and aggregation create a rapid plug at vascular injury.

Control architecture. Endothelium, von Willebrand factor, platelet receptors, shear, calcium, thromboxane and inhibitory mediators.

Failure modes. Bleeding from low/dysfunctional platelets or thrombosis from excessive activation.

Observable consequences. Platelet count, bleeding history, aggregation studies, closure tests, thrombus imaging and clinical events.

Cross-system coupling. Vascular biology, inflammation, drugs, liver disease, kidney disease and malignancy.

Evidence anchors: [2,3,6]

3.0.5 Coagulation and anticoagulant balance

Core principle. Protease cascades generate thrombin and fibrin while antithrombin, protein C/S, tissue factor pathway inhibitor, and fibrinolysis constrain clotting.

Control architecture. Tissue factor exposure, phospholipid surfaces, calcium, hepatic synthesis, vitamin K, flow and feedback loops.

Failure modes. Hemophilia, acquired factor deficiency, disseminated coagulation, venous/arterial thrombosis or failed fibrinolysis.

Observable consequences. PT/INR, aPTT, fibrinogen, factor assays, D-dimer, thrombin generation, viscoelastic tests and events.

Cross-system coupling. Liver, endothelium, pregnancy, infection, cancer, trauma and pharmacology.

Evidence anchors: [2,6,12]

3.0.6 Blood rheology and microcirculation

Core principle. Cell concentration, deformability, aggregation, plasma proteins, vessel caliber, and shear determine flow and tissue perfusion.

Control architecture. Hematocrit, temperature, red-cell mechanics, endothelial tone and local metabolic control.

Failure modes. Hyperviscosity, sickling, sludging, microthrombi, impaired capillary transit or hemorrhage.

Observable consequences. Viscosity, deformability, microvascular imaging, oxygen extraction, lactate and organ function.

Cross-system coupling. Cardiovascular, pulmonary, renal, neurologic and thermal systems.

Evidence anchors: [2,3,4]

3.0.7 Iron and micronutrient economy

Core principle. Iron is absorbed, transported, stored, recycled, and restricted during inflammation; folate and B12 support DNA synthesis.

Control architecture. Hepcidin-ferroportin axis, transferrin, ferritin, macrophage recycling, diet, absorption and losses.

Failure modes. Absolute or functional iron deficiency, overload, megaloblastosis or neurohematologic injury.

Observable consequences. Ferritin, transferrin saturation, soluble receptors, marrow iron, indices and response to therapy.

Cross-system coupling. GI absorption, liver storage, kidney disease, infection, pregnancy and cognition.

Evidence anchors: [3,7,11]

3.0.8 Transfusion compatibility and blood-component biology

Core principle. Donor red cells, platelets, plasma and specific factors can restore missing capacity but carry immunologic, infectious, volume and storage-related risks.

Control architecture. ABO/Rh and other antigens, antibody screening, component processing, storage, matching and hemovigilance.

Failure modes. Hemolysis, alloimmunization, transfusion reactions, overload, lung injury, infection or undertransfusion.

Observable consequences. Type/screen, crossmatch, reaction workup, component quality, clinical response and hemovigilance.

Cross-system coupling. Immunity, critical care, surgery, obstetrics, transplantation and health systems.

Evidence anchors: [6,7,16]

3.0.9 Clonal evolution and marrow ecology

Core principle. Somatic variants may expand with age or selection and can remain indolent, alter vascular risk, or progress to neoplasia.

Control architecture. Mutation, epigenetic state, inflammation, therapy, marrow niche and immune surveillance.

Failure modes. Clonal cytopenia, myelodysplasia, leukemia, myeloproliferation or treatment-resistant disease.

Observable consequences. Serial counts, morphology, cytogenetics, sequencing, allele fraction, marrow architecture and outcomes.

Cross-system coupling. Aging, cancer, inflammation, cardiovascular risk and precision medicine.

Evidence anchors: [3,8,17]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Arterial oxygen content	$C_{aO_2} = 1.34 Hb S_{aO_2} + 0.003 P_{aO_2}$	Hemoglobin-bound oxygen dominates content under ordinary conditions.	Constants and units vary; tissue delivery also depends on flow and extraction.
Corrected reticulocyte concept	$RPI \approx Retic\% \times \frac{Hct_{patient}}{Hct_{normal}} \div maturation\ factor$	Adjusts apparent reticulocyte fraction for anemia and prolonged maturation.	Approximate; analyzer-specific absolute reticulocyte counts and context are preferred.
INR	$INR = (PT_{patient} / PT_{mean\ normal})^{1.5}$	Standardizes prothrombin time for vitamin-K antagonist monitoring.	Not a universal measure of bleeding or coagulation and not valid for all drugs.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within hematologic health and hemostasis.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within hematologic health and hemostasis.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

Evidence anchors: [5,15]

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: ^[16]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the hematologic disorder, hemostatic failure mode, clonal state, or functional phenotype, but cross-system processes and functional effects are retained.

5.0.1 Iron-deficiency anemia

Definition and phenotype. Reduced hemoglobin production from insufficient available iron, usually reflecting loss, inadequate intake/absorption, or increased need.

Pathophysiology. Iron-restricted erythropoiesis produces small hypochromic cells and limits oxygen delivery before severe resting abnormalities appear.

Determinants and modifiers. Bleeding, menstruation, pregnancy, diet, malabsorption, inflammation and access to evaluation.

Assessment. CBC/indices, reticulocytes, ferritin interpreted with inflammation, transferrin saturation and investigation of cause.

Intervention logic. Correct cause; oral or intravenous iron selected by severity, tolerance and absorption; avoid treating the number without finding losses.

Known limits and open questions. Ferritin thresholds vary with inflammation and population; symptom attribution can be mixed.

Suite links. 03E GI; 03M reproductive; 03N maternal; 01 whole-person.

Evidence anchors: [6,11,18]

5.0.2 Anemia of inflammation and chronic disease

Definition and phenotype. Reduced erythropoiesis and iron availability during persistent immune, renal, malignant or systemic disease.

Pathophysiology. Cytokines raise hepcidin, shorten red-cell survival and blunt marrow response; kidney disease reduces erythropoietin.

Determinants and modifiers. Inflammatory burden, CKD, infection, cancer, age and nutrition.

Assessment. CBC, reticulocytes, iron studies, inflammatory context, renal function and exclusion of concurrent deficiencies.

Intervention logic. Treat underlying process; selected iron or erythropoiesis-stimulating therapy under indication-specific risk controls.

Known limits and open questions. Biomarkers poorly separate mixed absolute and functional deficiency in some settings.

Suite links. 03F renal; 03H immune; 04 inflammation.

Evidence anchors: [3,6,10]

5.0.3 Hemolytic anemia

Definition and phenotype. Premature red-cell destruction within vessels or reticuloendothelial organs from intrinsic or acquired causes.

Pathophysiology. Membrane, enzyme, hemoglobin, immune, mechanical, infectious or microangiopathic injury exceeds marrow compensation.

Determinants and modifiers. Inherited variants, autoimmunity, drugs, devices, infection, pregnancy and toxins.

Assessment. Reticulocytes, bilirubin, LDH, haptoglobin, smear, antiglobulin test and targeted molecular/enzymatic testing.

Intervention logic. Remove trigger; treat immune, infectious, mechanical or inherited mechanism; transfuse when net benefit warrants.

Known limits and open questions. Markers are context-dependent and transfusion can obscure phenotype.

Suite links. 03A cardiovascular devices; 03H immunity; 08 infection.

Evidence anchors: [2,3,6]

5.0.4 Sickle cell disease and related hemoglobinopathies

Definition and phenotype. Inherited hemoglobin disorders causing polymerization, hemolysis, vaso-occlusion and multisystem injury.

Pathophysiology. Deoxygenated abnormal hemoglobin changes cell mechanics, promotes adhesion/inflammation and obstructs microcirculation.

Determinants and modifiers. Genotype, modifiers, infection, dehydration, altitude, access to preventive care and structural inequity.

Assessment. Newborn screening, hemoglobin analysis/genotyping, organ surveillance, pain and function trajectories.

Intervention logic. Vaccination and infection prevention, disease-modifying therapy, transfusion strategies, curative transplantation/gene therapy in selected patients, and comprehensive pain care.

Known limits and open questions. Biomarkers do not fully predict crises; access and long-term durability of advanced therapies remain uneven.

Suite links. 03A, 03B, 03F, 03C, 10 rehabilitation, 15 equity.

Evidence anchors: [6,13,15]

5.0.5 Thalassemia syndromes

Definition and phenotype. Inherited imbalance of globin-chain production causing ineffective erythropoiesis, hemolysis and variable transfusion dependence.

Pathophysiology. Unpaired globin chains damage precursors and red cells; chronic marrow expansion and iron loading produce systemic injury.

Determinants and modifiers. Genotype, modifier alleles, transfusion access, chelation, infection control and reproductive screening.

Assessment. CBC/morphology, hemoglobin analysis, molecular testing, iron burden and organ surveillance.

Intervention logic. Transfusion and chelation when indicated; luspatercept or other disease-modifying options by phenotype; transplant/gene therapy in selected cases.

Known limits and open questions. Phenotype varies beyond genotype and long-term comparative evidence for newer therapies evolves.

Suite links. 03D endocrine; 03M reproductive; 14 therapeutics.

Evidence anchors: [6,7,15]

5.0.6 Platelet-number disorders

Definition and phenotype. Thrombocytopenia or thrombocytosis arising from production, destruction, sequestration, consumption, dilution or clonal expansion.

Pathophysiology. Platelet mass and function interact with vascular integrity and coagulation; count alone incompletely predicts bleeding or thrombosis.

Determinants and modifiers. Immune disease, infection, drugs, liver/spleen disease, pregnancy, critical illness and neoplasia.

Assessment. Repeat count excluding artifacts, smear, history, coagulation context, targeted immune/marrow/genetic studies.

Intervention logic. Treat cause and clinical risk; platelet transfusion or antiplatelet/cytoreductive strategies only under indication-specific thresholds.

Known limits and open questions. Risk thresholds vary by procedure and mechanism; pseudothrombocytopenia can mislead.

Suite links. 03H immune; 03N pregnancy; 04 sepsis; 14 pharmacology.

Evidence anchors: [6,11,12]

5.0.7 Inherited bleeding disorders

Definition and phenotype. Deficiencies or dysfunction of coagulation factors, von Willebrand factor or platelets causing mucocutaneous, deep-tissue or procedural bleeding.

Pathophysiology. Failure of platelet adhesion or fibrin generation destabilizes hemostasis according to pathway and residual activity.

Determinants and modifiers. Genotype, sex-linked inheritance, trauma/procedures, inhibitors and access to factor products.

Assessment. Structured bleeding history, PT/aPTT, factor/vWF assays, platelet studies and genetic testing when useful.

Intervention logic. Replacement, nonfactor agents, antifibrinolytics, desmopressin in selected states, procedural planning and inhibitor management.

Known limits and open questions. Laboratory activity and clinical bleeding can disagree; assay choice matters for novel agents.

Suite links. 03M/03N reproductive and obstetric care; 12 delivery systems.

Evidence anchors: [6,7]

5.0.8 Venous thromboembolism

Definition and phenotype. Deep-vein thrombosis and pulmonary embolism caused by interacting stasis, endothelial injury and hypercoagulability.

Pathophysiology. Coagulation activation and impaired fibrinolysis generate venous thrombus that may obstruct local flow or embolize.

Determinants and modifiers. Immobilization, surgery, cancer, pregnancy, estrogen, inherited thrombophilia, infection and prior events.

Assessment. Pretest probability, D-dimer in appropriate populations, compression ultrasound, CT/VQ imaging and hemodynamic severity.

Intervention logic. Anticoagulation selected by setting and risk; reperfusion or procedures for selected high-risk disease; prevention when net benefit supports it.

Known limits and open questions. Testing for thrombophilia is frequently overused and recurrence prediction remains imperfect.

Suite links. 03A cardiovascular; 03B respiratory; 03N maternal; 04 cancer/inflammation.

Evidence anchors: [2,6,11]

5.0.9 Disseminated intravascular coagulation and thrombotic microangiopathy

Definition and phenotype. Systemic coagulation or microvascular platelet-rich thrombosis with consumption and organ injury from distinct initiating mechanisms.

Pathophysiology. Tissue-factor/coagulation activation, endothelial injury, complement or ADAMTS13-related failure produce thrombosis with bleeding risk.

Determinants and modifiers. Sepsis, trauma, obstetric catastrophe, cancer, drugs, autoimmune and genetic conditions.

Assessment. Serial counts, smear, coagulation, fibrin markers, hemolysis, organ injury and mechanism-specific testing.

Intervention logic. Urgently treat cause; supportive blood components by bleeding/risk; plasma exchange, complement inhibition or other targeted therapy where indicated.

Known limits and open questions. No single test establishes all forms; timing and phenotype overlap.

Suite links. 04 cross-system processes; 03H immunity; 03N obstetrics; 12 critical care.

Evidence anchors: [6,11,12]

5.0.10 Bone-marrow failure syndromes

Definition and phenotype. Inadequate production of one or more blood lineages from immune, inherited, toxic, infectious or clonal injury.

Pathophysiology. Stem/progenitor depletion or ineffective hematopoiesis produces cytopenias and vulnerability to infection, bleeding and hypoxia.

Determinants and modifiers. Genetics, age, immune attack, radiation/chemicals, drugs, viruses and prior therapy.

Assessment. Serial counts, reticulocytes, marrow morphology/cellularity, cytogenetics, flow and molecular testing.

Intervention logic. Remove cause; supportive care; immunosuppression, growth factors or transplantation according to mechanism and severity.

Known limits and open questions. Early clonal disease can be difficult to distinguish from reactive or immune failure.

Suite links. 08 infection; 09 occupational exposure; 14 therapies.

Evidence anchors: [3,6,8]

5.0.11 Myelodysplastic and myeloproliferative neoplasms

Definition and phenotype. Clonal disorders with ineffective production, dysplasia, overproduction, fibrosis, thrombosis and variable progression.

Pathophysiology. Somatic driver mutations reshape stem-cell fitness, differentiation, inflammation and marrow architecture.

Determinants and modifiers. Age, prior cytotoxic exposure, inherited predisposition and stochastic clonal evolution.

Assessment. Counts, smear, marrow, cytogenetics, molecular profile, symptoms, thrombosis and prognostic models.

Intervention logic. Observation, supportive care, cytoreduction, targeted agents, hypomethylating therapy or transplantation by risk and goals.

Known limits and open questions. Molecular complexity and competing risks limit one-size staging; treatment can alter clonal selection.

Suite links. 03A thrombosis; 14 oncology/pharmacology; 07 aging.

Evidence anchors: [6,8,17]

5.0.12 Acute and chronic leukemias

Definition and phenotype. Malignant expansion of hematopoietic cells that disrupts marrow, immunity and organs with highly variable tempo.

Pathophysiology. Genetic/epigenetic lesions block differentiation, increase survival or drive proliferation; microenvironment and therapy select resistant clones.

Determinants and modifiers. Age, inherited predisposition, prior therapy/radiation and specific exposures, with many cases lacking a known preventable cause.

Assessment. Blood and marrow morphology, flow, cytogenetics/molecular classification, disease burden and organ assessment.

Intervention logic. Subtype-specific combinations of cytotoxic, targeted, immune and cellular therapies; transplantation and supportive care where appropriate.

Known limits and open questions. Rapidly evolving classifications and minimal-residual-disease assays require method-specific interpretation.

Suite links. 14 therapeutics; 03H immune; 08 infection; 15 access.

Evidence anchors: [6,8,15]

5.0.13 Lymphoma and plasma-cell disorders

Definition and phenotype. Clonal lymphoid or plasma-cell diseases involving nodes, marrow, blood, protein production and organs.

Pathophysiology. Aberrant antigen-receptor biology, signaling, genomic lesions and microenvironmental support drive proliferation and immune dysfunction.

Determinants and modifiers. Age, immune deficiency, selected infections, inherited predisposition and exposures.

Assessment. Tissue diagnosis, immunophenotype, molecular studies, imaging, marrow, monoclonal-protein assays and organ injury.

Intervention logic. Risk-adapted systemic therapy, targeted/immune/cellular treatment, radiation or observation depending on subtype and burden.

Known limits and open questions. Indolent disease, precursor states and treatment toxicity complicate thresholds.

Suite links. 03H immune; 03F renal; 14 cancer therapy.

Evidence anchors: [6,8,9]

5.0.14 Transfusion-related and alloimmune disorders

Definition and phenotype. Adverse outcomes caused by immune incompatibility, component characteristics, volume, storage, contamination or repeated exposure.

Pathophysiology. Antibody-mediated hemolysis, leukocyte reactions, lung endothelial injury, overload and infectious or metabolic complications.

Determinants and modifiers. Prior pregnancy/transfusion, immune state, component type, rapid volume, critical illness and system quality.

Assessment. Bedside recognition, clerical check, hemolysis/antibody testing, cultures, imaging and hemovigilance classification.

Intervention logic. Stop implicated component, stabilize, investigate, and modify future matching/component strategy; prevention through system design.

Known limits and open questions. Symptoms overlap underlying illness and reactions are underreported.

Suite links. 12 safety systems; 03A/03B overload and lung injury; 03N obstetrics.

Evidence anchors: ^[6,7,16]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current hematologic health and hemostasis risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.

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Stage	Purpose	Components	Output	Common failure
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

6.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current hematologic health and hemostasis risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

6.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.

Evidence anchors: ^[27]

6.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.

Evidence anchors: ^[3]

6.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

6.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.

6.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Evidence anchors: ^[25]

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

Measurement is an inference chain



Validation must test every link, including population, setting, motion, missingness, drift, and downstream use.

Figure 7.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Complete blood count and indices	Noninvasive venous/capillary sample	Impedance or optical cell counting, hemoglobin photometry and derived indices.	Lineage counts, indices, flags and distributions.	Fast, scalable first-line map.	Preanalytic dilution/clotting, instrument flags, reference-range and morphology limitations.
Peripheral blood smear	Minimally invasive sample	Stained-cell optical morphology interpreted manually or digitally.	Cell size, shape, inclusions, parasites, blasts, fragmentation and platelet estimate.	Direct structural context and rapid pattern recognition.	Operator dependence, sampling, staining and limited quantitation.
Reticulocyte and immature-cell analysis	Minimally invasive sample	Fluorescent nucleic-acid staining and flow/optical counting.	Marrow response, immature fractions and production indices.	Separates underproduction from compensation.	Affected by transfusion, assay definitions and extreme counts.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Coagulation assays	Minimally invasive sample	Clot time, chromogenic, immunologic and viscoelastic responses after controlled reagents.	PT/INR, aPTT, fibrinogen, factor activity, anti-Xa and global clot dynamics.	Mechanism-targeted monitoring and diagnosis.	Preanalytics, reagent sensitivity, inhibitors, drug interference and non-equivalence.
Flow cytometry	Minimally invasive to invasive sample	Fluorophore-labeled antibodies and light scatter on single cells.	Lineage, activation, clonality, rare cells and residual disease.	High-dimensional cellular phenotyping.	Panel design, compensation, gating, specimen age and interlaboratory variability.
Bone marrow aspiration and biopsy	Invasive	Cell aspirate and core architecture examined by morphology, flow, cytogenetics and molecular assays.	Cellularity, lineage maturation, fibrosis, infiltration, storage and clonal disease.	Direct assessment of hematopoietic tissue.	Pain, sampling error, dry tap, focal disease and interpretive complexity.
Hemoglobin analysis and red-cell functional tests	Minimally invasive sample	Electrophoresis/HPLC/capillary electrophoresis, molecular assays, enzyme, osmotic and deformability tests.	Hemoglobin fractions, variants and intrinsic red-cell defects.	Mechanism-specific characterization.	Transfusion, age, co-inherited variants and method-specific blind spots.
Iron, vitamin and hemolysis biomarkers	Minimally invasive sample	Photometric, immunoassay and mass-spectrometric chemistry.	Ferritin, transferrin saturation, B12/folate, bilirubin, LDH, haptoglobin and related markers.	Scalable mechanism clues and monitoring.	Inflammation, liver/kidney disease and assay interference reduce specificity.
Cytogenetics, sequencing and clonality assays	Minimally invasive to invasive sample	Karyotype, FISH, PCR, next-generation sequencing and allele quantification.	Drivers, prognosis, targetability, inherited predisposition and residual disease.	High specificity and longitudinal clonal tracking.	Incidental germline findings, low-level clones, platform limits and uncertain variants.
Blood-bank immunohematology	Minimally invasive sample	Agglutination, solid-phase/gel, antibody identification and molecular typing.	ABO/Rh, alloantibodies, compatibility and reaction workup.	Prevents catastrophic incompatibility.	Rare antigens, recent transfusion, weak antibodies, clerical error and supply constraints.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Thromboelastography/thromboelastometry	Minimally invasive sample; bedside	Mechanical or optical detection of clot initiation, strength and lysis in whole blood.	Global viscoelastic curves and treatment-oriented parameters.	Rapid integrated view in bleeding settings.	Device/reagent-specific values, limited representation of endothelium and uncertain benefit outside pathways.
Vascular imaging for thrombosis	Noninvasive to minimally invasive contrast imaging	Compression ultrasound, CT/MR angiography or ventilation-perfusion methods.	Thrombus location, obstruction, embolic burden and alternative diagnoses.	Anatomic decision support.	Operator/contrast/radiation constraints and residual clot ambiguity.
Point-of-care hematology and coagulation	Minimally invasive bedside/home	Microfluidic, electrochemical, optical or mechanical cartridge systems.	Selected counts, hemoglobin, INR, blood type or coagulation measures.	Fast decentralized decisions.	Small-volume sampling, hematocrit effects, temperature, calibration and user errors.
Microfluidic cell mechanics and clot models	Ex vivo research	Controlled microchannels, imaging and force/flow transduction.	Deformability, adhesion, occlusion, thrombus kinetics and drug response.	Mechanistic and low-volume phenotyping.	Device geometry, surface chemistry and ex vivo conditions may not reproduce vessels.
Continuous or minimally invasive blood sensing	Minimally invasive/implantable research	Optical, electrochemical, acoustic or affinity sensing of cells, coagulation or chemistry over time.	Trends in hemoglobin, viscosity, clot risk or biomarkers.	Potential temporal resolution beyond episodic blood draws.	Biofouling, drift, specificity, calibration, thrombosis and infection.

7.0.1 Complete blood count and indices

Measured quantity and transducer. Impedance or optical cell counting, hemoglobin photometry and derived indices.

Typical outputs. Lineage counts, indices, flags and distributions.

Strengths. Fast, scalable first-line map.

Limitations and artifacts. Preanalytic dilution/clotting, instrument flags, reference-range and morphology limitations.

Reference and validation strategy. Traceable controls, smear review rules, repeatability and clinical correlation.

Clinical/research maturity. Established laboratory standard.

Evidence anchors: [6,7]

7.0.2 Peripheral blood smear

Measured quantity and transducer. Stained-cell optical morphology interpreted manually or digitally.

Typical outputs. Cell size, shape, inclusions, parasites, blasts, fragmentation and platelet estimate.

Strengths. Direct structural context and rapid pattern recognition.

Limitations and artifacts. Operator dependence, sampling, staining and limited quantitation.

Reference and validation strategy. Blinded expert comparison, digital calibration and proficiency testing.

Clinical/research maturity. Established; AI assistance emerging.

Evidence anchors: [3,20]

7.0.3 Reticulocyte and immature-cell analysis

Measured quantity and transducer. Fluorescent nucleic-acid staining and flow/optical counting.

Typical outputs. Marrow response, immature fractions and production indices.

Strengths. Separates underproduction from compensation.

Limitations and artifacts. Affected by transfusion, assay definitions and extreme counts.

Reference and validation strategy. Reference materials, method comparison and clinical-state validation.

Clinical/research maturity. Established.

Evidence anchors: [6]

7.0.4 Coagulation assays

Measured quantity and transducer. Clot time, chromogenic, immunologic and viscoelastic responses after controlled reagents.

Typical outputs. PT/INR, aPTT, fibrinogen, factor activity, anti-Xa and global clot dynamics.

Strengths. Mechanism-targeted monitoring and diagnosis.

Limitations and artifacts. Preanalytics, reagent sensitivity, inhibitors, drug interference and non-equivalence.

Reference and validation strategy. Calibrators, proficiency, drug-specific and outcome-linked validation.

Clinical/research maturity. Established with assay-specific limits.

Evidence anchors: [6,21]

7.0.5 Flow cytometry

Measured quantity and transducer. Fluorophore-labeled antibodies and light scatter on single cells.

Typical outputs. Lineage, activation, clonality, rare cells and residual disease.

Strengths. High-dimensional cellular phenotyping.

Limitations and artifacts. Panel design, compensation, gating, specimen age and interlaboratory variability.

Reference and validation strategy. Standardized controls, harmonized gating, external proficiency and molecular/tissue correlation.

Clinical/research maturity. Established specialist platform.

Evidence anchors: [6,8]

7.0.6 Bone marrow aspiration and biopsy

Measured quantity and transducer. Cell aspirate and core architecture examined by morphology, flow, cytogenetics and molecular assays.

Typical outputs. Cellularity, lineage maturation, fibrosis, infiltration, storage and clonal disease.

Strengths. Direct assessment of hematopoietic tissue.

Limitations and artifacts. Pain, sampling error, dry tap, focal disease and interpretive complexity.

Reference and validation strategy. Adequate core/aspirate, integrated pathology and repeat sampling when discordant.

Clinical/research maturity. Established invasive reference.

Evidence anchors: [6,8]

7.0.7 Hemoglobin analysis and red-cell functional tests

Measured quantity and transducer. Electrophoresis/HPLC/capillary separation, molecular assays, enzyme, osmotic and deformability tests.

Typical outputs. Hemoglobin fractions, variants and intrinsic red-cell defects.

Strengths. Mechanism-specific characterization.

Limitations and artifacts. Transfusion, age, co-inherited variants and method-specific blind spots.

Reference and validation strategy. Orthogonal confirmation and genotype-phenotype correlation.

Clinical/research maturity. Established to advanced.

Evidence anchors: [6,15]

7.0.8 Iron, vitamin and hemolysis biomarkers

Measured quantity and transducer. Photometric, immunoassay and mass-spectrometric chemistry.

Typical outputs. Ferritin, transferrin saturation, B12/folate, bilirubin, LDH, haptoglobin and related markers.

Strengths. Scalable mechanism clues and monitoring.

Limitations and artifacts. Inflammation, liver/kidney disease and assay interference reduce specificity.

Reference and validation strategy. Traceability, contextual panels and response-to-treatment validation.

Clinical/research maturity. Established.

Evidence anchors: [3,10]

7.0.9 Cytogenetics, sequencing and clonality assays

Measured quantity and transducer. Karyotype, FISH, PCR, next-generation sequencing and allele quantification.

Typical outputs. Drivers, prognosis, targetability, inherited predisposition and residual disease.

Strengths. High specificity and longitudinal clonal tracking.

Limitations and artifacts. Incidental germline findings, low-level clones, platform limits and uncertain variants.

Reference and validation strategy. Orthogonal confirmation, controls, longitudinal proficiency and clinical utility studies.

Clinical/research maturity. Established in defined indications; expanding.

Evidence anchors: [8,15,25]

7.0.10 Blood-bank immunohematology

Measured quantity and transducer. Agglutination, solid-phase/gel, antibody identification and molecular typing.

Typical outputs. ABO/Rh, alloantibodies, compatibility and reaction workup.

Strengths. Prevents catastrophic incompatibility.

Limitations and artifacts. Rare antigens, recent transfusion, weak antibodies, clerical error and supply constraints.

Reference and validation strategy. End-to-end traceability, external proficiency and hemovigilance.

Clinical/research maturity. Established safety-critical system.

Evidence anchors: [7,16]

7.0.11 Thromboelastography/rotational thromboelastometry

Measured quantity and transducer. Mechanical or optical detection of clot initiation, strength and lysis in whole blood.

Typical outputs. Global viscoelastic curves and treatment-oriented parameters.

Strengths. Rapid integrated view in bleeding settings.

Limitations and artifacts. Device/reagent-specific values, limited representation of endothelium and uncertain benefit outside pathways.

Reference and validation strategy. Method-specific reference, protocol trials and outcome validation.

Clinical/research maturity. Established in selected surgery/trauma; context-limited elsewhere.

Evidence anchors: [6,16]

7.0.12 Vascular imaging for thrombosis

Measured quantity and transducer. Compression ultrasound, CT/MR angiography or ventilation-perfusion methods.

Typical outputs. Thrombus location, obstruction, embolic burden and alternative diagnoses.

Strengths. Anatomic decision support.

Limitations and artifacts. Operator/contrast/radiation constraints and residual clot ambiguity.

Reference and validation strategy. Standard protocols, blinded interpretation and outcome-linked diagnostic pathways.

Clinical/research maturity. Established.

Evidence anchors: [2,6]

7.0.13 Point-of-care hematology and coagulation

Measured quantity and transducer. Microfluidic, electrochemical, optical or mechanical cartridge systems.

Typical outputs. Selected counts, hemoglobin, INR, blood type or coagulation measures.

Strengths. Fast decentralized decisions.

Limitations and artifacts. Small-volume sampling, hematocrit effects, temperature, calibration and user errors.

Reference and validation strategy. Bench interference, usability, lot, environmental and clinical method comparison.

Clinical/research maturity. Established for narrow uses; research expanding.

Evidence anchors: [19,21,24]

7.0.14 Microfluidic cell mechanics and clot models

Measured quantity and transducer. Controlled microchannels, imaging and force/flow transduction.

Typical outputs. Deformability, adhesion, occlusion, thrombus kinetics and drug response.

Strengths. Mechanistic and low-volume phenotyping.

Limitations and artifacts. Device geometry, surface chemistry and ex vivo conditions may not reproduce vessels.

Reference and validation strategy. Reference materials, cross-platform studies and prospective clinical utility.

Clinical/research maturity. Research/frontier.

7.0.15 Continuous or minimally invasive blood sensing

Measured quantity and transducer. Optical, electrochemical, acoustic or affinity sensing of cells, coagulation or chemistry over time.

Typical outputs. Trends in hemoglobin, viscosity, clot risk or biomarkers.

Strengths. Potential temporal resolution beyond episodic blood draws.

Limitations and artifacts. Biofouling, drift, specificity, calibration, thrombosis and infection.

Reference and validation strategy. Long-duration bench and animal work before controlled human reference studies.

Clinical/research maturity. Frontier.

Evidence anchors: ^[19,22,23]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise hematologic health and hemostasis quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

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Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

8.0.1 Intended use and measurand

Design requirements. Define the precise hematologic health and hemostasis quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

Evidence anchors: [19,22]

8.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

Evidence anchors: [22,23]

8.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

Evidence anchors: ^[25]

8.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

Evidence anchors: ^[21]

8.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

Evidence anchors: ^[21,25]

8.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

8.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: [20]

8.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Evidence anchors: [23,24]

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Nutrition and cause-directed deficiency correction	Prevention/treatment	Restoration required substrates and stop blood loss or malabsorption.	Strong for defined deficiencies when cause and dose are appropriate.	Overtreatment, iron overload, missed malignancy/bleeding and access barriers.	Counts, iron/vitamin status, symptoms, source control and adverse effects.
Transfusion and component therapy	Acute/supportive	Replace red-cell oxygen capacity, platelets, plasma, fibrinogen or specific components.	Life-saving in selected bleeding, severe cytopenia and procedural states; restrictive strategies often reduce exposure.	Reaction, overload, lung injury, alloimmunization, infection, scarcity and inequity.	Indication, dose, clinical response, reaction surveillance and hemovigilance.
Anticoagulant therapy	Treatment/prevention	Reduction of thrombin generation or factor activity to prevent/treat thrombosis.	Strong in defined venous, cardiac and procedural indications.	Bleeding, interactions, renal/hepatic accumulation, adherence and peri-procedural complexity.	Renal/liver function, interactions, bleeding/thrombosis, drug-specific tests when indicated.
Antiplatelet therapy	Treatment/prevention	Inhibition of platelet activation/aggregation for selected arterial and device-related risk.	Strong in defined cardiovascular settings; not interchangeable with anticoagulation.	Bleeding, resistance/nonadherence and inappropriate extension to low-risk people.	Clinical events, bleeding, adherence and procedure planning.
Hemostatic replacement and antifibrinolysis	Acute/chronic	Replace deficient factors or stabilize clot in bleeding disorders and procedures.	High efficacy when matched to mechanism.	Thrombosis, inhibitors, assay interference and product access.	Factor/drug levels where relevant, bleeding control and thrombosis.

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Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Cytoreductive and disease-modifying hematology therapy	Chronic/specialist	Reduce abnormal clones or modify red-cell/platelet phenotype.	Outcome benefit depends on disease, molecular risk and treatment goal.	Myelosuppression, thrombosis, infection, teratogenicity and secondary malignancy.	Counts, molecular burden, organ function, symptoms and adverse events.
Targeted, immune and cellular therapy	Specialist/potentially curative	Exploit molecular lesions or immune recognition in malignant and selected nonmalignant disease.	Transformative in defined subtypes; response and durability vary.	Cytokine toxicity, neurotoxicity, immune suppression, resistance, cost and access.	Disease-specific response, residual disease, immune toxicity, infection and long-term follow-up.
Hematopoietic stem-cell transplantation	Potentially curative	Replace diseased hematopoiesis and/or provide graft-versus-tumor effect.	Curative potential for selected malignant, marrow-failure and inherited disorders.	Mortality, graft-versus-host disease, infection, infertility, organ injury and access.	Donor/chimerism, disease burden, immune recovery, organ and functional outcomes.
Gene addition/editing and ex vivo cell engineering	Potentially curative/frontier	Correct or compensate inherited defects in autologous stem cells.	Strong efficacy for selected disorders with evolving durability data.	Conditioning toxicity, off-target/clonal risk, manufacturing failure, cost and global inequity.	Molecular correction, blood phenotype, clonal surveillance, organ outcomes and decades-long follow-up.
Infection prevention and supportive care	Prevention/supportive	Vaccination, antimicrobial prophylaxis where indicated, fever pathways and growth-factor/supportive strategies.	Reduces preventable morbidity in severe cytopenia and functional asplenia.	Resistance, adverse effects and unequal access.	Vaccination, infection events, antimicrobial exposure and marrow recovery.
Patient blood management	System intervention	Prevent avoidable anemia/bleeding, optimize coagulation and use components appropriately.	Can reduce exposure, complications and resource waste when implemented as a pathway.	Protocol rigidity or undertreatment if detached from clinical context.	Transfusion rates, bleeding, outcomes, appropriateness, equity and adverse events.

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Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Rehabilitation, pain and survivorship care	Whole-person	Restore function and manage fatigue, pain, cognition, work, fertility and late effects.	Important for chronic hemoglobinopathy, cancer and treatment survivorship despite variable trial depth.	Fragmented access and underrecognition of disability.	PROMs, activity, participation, return to roles, symptoms and goals.

9.0.1 Nutrition and cause-directed deficiency correction

Target and mechanism. Restore required substrates and stop blood loss or malabsorption.

Evidence and expected outcome. Strong for defined deficiencies when cause and dose are appropriate.

Risks, burden, and equity. Overtreatment, iron overload, missed malignancy/bleeding and access barriers.

Response monitoring. Counts, iron/vitamin status, symptoms, source control and adverse effects.

Evidence anchors: [6,18]

9.0.2 Transfusion and component therapy

Target and mechanism. Replace red-cell oxygen capacity, platelets, plasma, fibrinogen or specific components.

Evidence and expected outcome. Life-saving in selected bleeding, severe cytopenia and procedural states; restrictive strategies often reduce exposure.

Risks, burden, and equity. Reaction, overload, lung injury, alloimmunization, infection, scarcity and inequity.

Response monitoring. Indication, dose, clinical response, reaction surveillance and hemovigilance.

Evidence anchors: [7,16]

9.0.3 Anticoagulant therapy

Target and mechanism. Reduce thrombin generation or factor activity to prevent/treat thrombosis.

Evidence and expected outcome. Strong in defined venous, cardiac and procedural indications.

Risks, burden, and equity. Bleeding, interactions, renal/hepatic accumulation, adherence and peri-procedural complexity.

Response monitoring. Renal/liver function, interactions, bleeding/thrombosis, drug-specific tests when indicated.

Evidence anchors: [2,6]

9.0.4 Antiplatelet therapy

Target and mechanism. Inhibit platelet activation/aggregation for selected arterial and device-related risk.

Evidence and expected outcome. Strong in defined cardiovascular settings; not interchangeable with anticoagulation.

Risks, burden, and equity. Bleeding, resistance/nonadherence and inappropriate extension to low-risk people.

Response monitoring. Clinical events, bleeding, adherence and procedure planning.

Evidence anchors: [2,6]

9.0.5 Hemostatic replacement and antifibrinolysis

Target and mechanism. Replace deficient factors or stabilize clot in bleeding disorders and procedures.

Evidence and expected outcome. High efficacy when matched to mechanism.

Risks, burden, and equity. Thrombosis, inhibitors, assay interference and product access.

Response monitoring. Factor/drug levels where relevant, bleeding control and thrombosis.

Evidence anchors: [6,7]

9.0.6 Cytoreductive and disease-modifying hematology therapy

Target and mechanism. Reduce abnormal clones or modify red-cell/platelet phenotype.

Evidence and expected outcome. Outcome benefit depends on disease, molecular risk and treatment goal.

Risks, burden, and equity. Myelosuppression, thrombosis, infection, teratogenicity and secondary malignancy.

Response monitoring. Counts, molecular burden, organ function, symptoms and adverse events.

Evidence anchors: [6,8]

9.0.7 Targeted, immune and cellular therapy

Target and mechanism. Exploit molecular lesions or immune recognition in malignant and selected non-malignant disease.

Evidence and expected outcome. Transformative in defined subtypes; response and durability vary.

Risks, burden, and equity. Cytokine toxicity, neurotoxicity, immune suppression, resistance, cost and access.

Response monitoring. Disease-specific response, residual disease, immune toxicity, infection and long-term follow-up.

Evidence anchors: [8,9,15]

9.0.8 Hematopoietic stem-cell transplantation

Target and mechanism. Replace diseased hematopoiesis and/or provide graft-versus-tumor effect.

Evidence and expected outcome. Curative potential for selected malignant, marrow-failure and inherited disorders.

Risks, burden, and equity. Mortality, graft-versus-host disease, infection, infertility, organ injury and access.

Response monitoring. Donor/chimerism, disease burden, immune recovery, organ and functional outcomes.

Evidence anchors: [6,8]

9.0.9 Gene addition/editing and ex vivo cell engineering

Target and mechanism. Correct or compensate inherited defects in autologous stem cells.

Evidence and expected outcome. Strong efficacy for selected disorders with evolving durability data.

Risks, burden, and equity. Conditioning toxicity, off-target/clonal risk, manufacturing failure, cost and global inequity.

Response monitoring. Molecular correction, blood phenotype, clonal surveillance, organ outcomes and decades-long follow-up.

Evidence anchors: [8,15]

9.0.10 Infection prevention and supportive care

Target and mechanism. Vaccination, antimicrobial prophylaxis where indicated, fever pathways and growth-factor/supportive strategies.

Evidence and expected outcome. Reduces preventable morbidity in severe cytopenia and functional asplenia.

Risks, burden, and equity. Resistance, adverse effects and unequal access.

Response monitoring. Vaccination, infection events, antimicrobial exposure and marrow recovery.

Evidence anchors: [6,13,14]

9.0.11 Patient blood management

Target and mechanism. Prevent avoidable anemia/bleeding, optimize coagulation and use components appropriately.

Evidence and expected outcome. Can reduce exposure, complications and resource waste when implemented as a pathway.

Risks, burden, and equity. Protocol rigidity or undertreatment if detached from clinical context.

Response monitoring. Transfusion rates, bleeding, outcomes, appropriateness, equity and adverse events.

Evidence anchors: [7,16]

9.0.12 Rehabilitation, pain and survivorship care

Target and mechanism. Restore function and manage fatigue, pain, cognition, work, fertility and late effects.

Evidence and expected outcome. Important for chronic hemoglobinopathy, cancer and treatment survivorship despite variable trial depth.

Risks, burden, and equity. Fragmented access and underrecognition of disability.

Response monitoring. PROMs, activity, participation, return to roles, symptoms and goals.

Evidence anchors: [26,27,28]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Can a global clotting assay predict individual bleeding or thrombosis?	Global assays summarize selected ex vivo clot kinetics.	Incremental clinical utility across heterogeneous diseases and therapies.	Endothelium, flow and tissue context are incompletely represented.	Prospective decision-impact studies with clinical outcomes and prespecified thresholds.
How should clonal hematopoiesis alter prevention?	Age-related clones associate with malignancy and some inflammatory vascular outcomes.	Who benefits from surveillance or targeted intervention.	Absolute risks vary and no established clone-directed prevention applies broadly.	Longitudinal subtype cohorts and mechanism-linked randomized interventions.
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within hematologic health and hemostasis.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

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Question	What is established	What remains uncertain	Why unresolved	Decisive next step
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

10.0.1 Can a global clotting assay predict individual bleeding or thrombosis?

Established knowledge. Global assays summarize selected ex vivo clot kinetics.

Unresolved issue. Incremental clinical utility across heterogeneous diseases and therapies.

Why current evidence is insufficient. Endothelium, flow and tissue context are incompletely represented.

Research that would change confidence. Prospective decision-impact studies with clinical outcomes and prespecified thresholds.

Evidence anchors: ^[6]

10.0.2 How should clonal hematopoiesis alter prevention?

Established knowledge. Age-related clones associate with malignancy and some inflammatory vascular outcomes.

Unresolved issue. Who benefits from surveillance or targeted intervention.

Why current evidence is insufficient. Absolute risks vary and no established clone-directed prevention applies broadly.

Research that would change confidence. Longitudinal subtype cohorts and mechanism-linked randomized interventions.

Evidence anchors: ^[2,17]

10.0.3 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within hematologic health and hemostasis.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

Evidence anchors: ^[15]

10.0.4 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: ^[27]

10.0.5 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

Evidence anchors: ^[8]

10.0.6 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

Evidence anchors: ^[19]

10.0.7 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Evidence anchors: ^[27]

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Multiscale hemostasis reference atlas	Can cellular, plasma and flow measures explain bleeding/thrombosis better than routine tests?	Prospective cohort with standardized sampling, microfluidics, global and conventional assays, endothelial context and adjudicated outcomes.	Externally validated improvement in decisions and calibration.	High-dimensional association without utility.
0. Safe continuous hemoglobin platform	Can trend sensing detect clinically important change without repeated phlebotomy?	Bench optical/tissue phantom work, controlled blood-loss/hemodilution studies, subgroup and perfusion challenges, blinded clinical comparison.	Prespecified individual error and actionable trend performance.	Correlation driven by population range or failure in low perfusion/skin subgroups.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for hematologic health and hemostasis?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.

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Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

11.0.1 0. Multiscale hemostasis reference atlas

Question. Can cellular, plasma and flow measures explain bleeding/thrombosis better than routine tests?

Design. Prospective cohort with standardized sampling, microfluidics, global and conventional assays, endothelial context and adjudicated outcomes.

Success criterion. Externally validated improvement in decisions and calibration.

Failure or stop criterion. High-dimensional association without utility.

Evidence anchors: [6,25]

11.0.2 0. Safe continuous hemoglobin platform

Question. Can trend sensing detect clinically important change without repeated phlebotomy?

Design. Bench optical/tissue phantom work, controlled blood-loss/hemodilution studies, subgroup and perfusion challenges, blinded clinical comparison.

Success criterion. Prespecified individual error and actionable trend performance.

Failure or stop criterion. Correlation driven by population range or failure in low perfusion/skin subgroups.

Evidence anchors: [19,22]

11.0.3 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for hematologic health and hemostasis?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

Evidence anchors: ^[25]

11.0.4 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

11.0.5 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: ^[22]

11.0.6 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

11.0.7 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

11.0.8 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
03A	Cardiovascular Health	Blood cells, thrombosis, bleeding, viscosity and devices directly alter circulation.	interpreting oxygen delivery, embolism, antithrombotic therapy or mechanical support.
03B	Respiratory Health	Hemoglobin and pulmonary gas exchange jointly determine arterial oxygen content.	saturation and symptoms disagree or pulmonary embolism is suspected.
03H	Immune and Lymphatic Health	Leukocytes, antibodies, complement, inflammation and immune therapies cross both domains.	cytopenia, autoimmunity, infection or immune malignancy is present.
04	Cross-System Pathological Processes	Sepsis, inflammation, cancer, shock and microvascular injury drive systemic hematologic failure.	coagulopathy is part of multisystem illness.
14	Interventions and Therapeutics	Anticoagulants, transfusion, cytotoxic, targeted, cellular and gene therapies require shared safety models.	moving from assay or classification to treatment.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Hematopoiesis	Production and maturation of blood cells from stem and progenitor cells.
Hemostasis	Coordinated vascular, platelet, coagulation and fibrinolytic processes that limit blood loss.
Cytopenia	Reduction in one or more circulating blood-cell lineages.
Clonal hematopoiesis	Expansion of blood-cell clones carrying acquired variants, without necessarily meeting criteria for malignancy.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

- [1] Technical Evidence and Systems Review. Atlas of Human Health: A Unified Taxonomy. 2026. Master taxonomy and conceptual architecture for this suite.
- [2] Technical Evidence and Engineering Review. Cardiovascular Health and Measurement Engineering. 2026. Volume 03A of this suite.
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Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.